Worksheet 4.10

Chemical patterns: Summary

Answers

Complete the statements below to compile a summary of this unit. The missing words can be found in the word list below.

1. All elements contain only one type of ….atom…. .
2. The central part of the atom is called the ..nucleus…, which consists of particles called .protons…. and …neutrons.. .
3. The nucleus is surrounded by a lot of empty space and a cloud of ..electrons.. .
4. Protons have a ..positive… charge, electrons have a ..negative… charge and neutrons have …..no……. charge.
5. The classification system used to order the elements is called the ..periodic… table.
6. The number of protons in an atom is referred to as the …atomic…. number.
7. The number of protons and neutrons in an atom is referred to as the …mass…… number.
8. Isotopes…. are different forms of atoms of the same element. Those of the same element have the same number of protons but a different number of neutrons.
9. The .metalloids.. have properties intermediate between that of metals and non-metals.
10. The energy levels in which electrons are arranged are called ….shells…. .
11. Atoms that have lost or gained an electron and therefore carry an electric charge are called …..ions…… .
12. When electrons are transferred from one atom to another, an ….ionic…… bond forms.
13. When electrons are shared between atoms, a .covalent… bond forms.
14. The number of electrons that an atom loses, gains or shares with other atoms is called its …valency… .
15. The shorthand way of writing an element's name is called its ..symbol…. .
16. The shorthand way of writing a compound's name is called its ..formula… .

periodic	atomic	symbol	mass
protons	nucleus	metalloids	formula
shells	isotopes	valency	no
negative	covalent	neutrons	electrons
ions	atom	positive	ionic

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